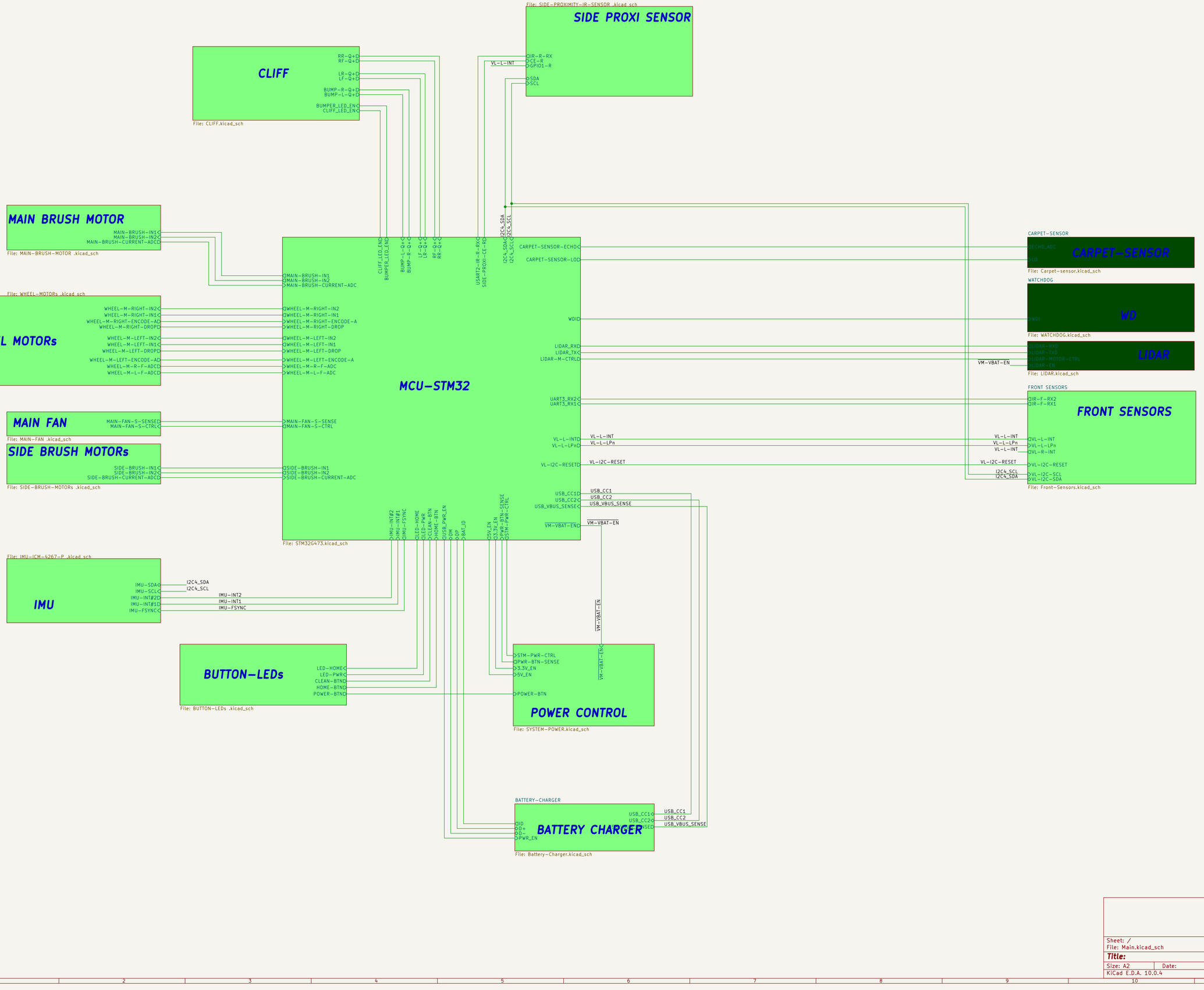
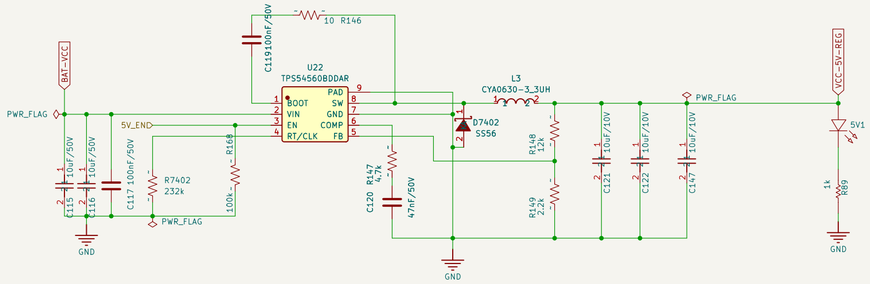


Sheet: /
File: Main.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/16

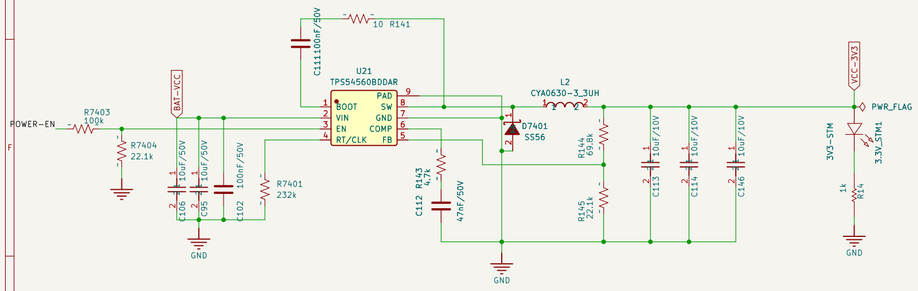
POWER MANAGMENT

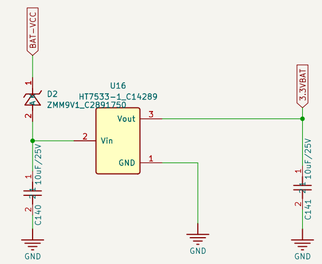


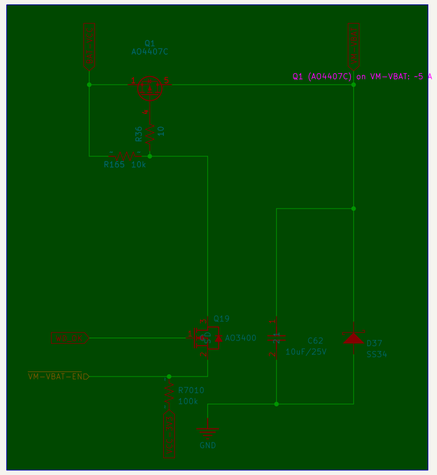
PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.



TPS54560B (U21, U22): non-synchronous,
SS56 catch diode; fsw ≈18 kHz (RT 232k);
COMP 4.7k + 47nF (all-ceramic Cout, no pole cap).
EN abs max 8.4 V; R7403/R7404 on U21.



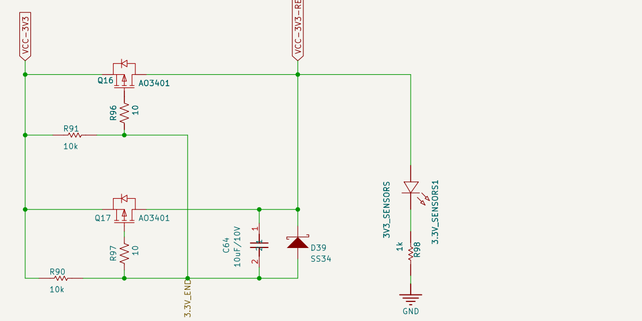


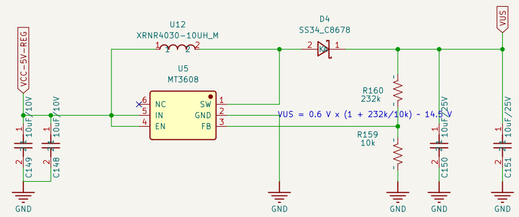


Q1 (AO4407C) on VM-VBAT: ~5 A normal load is ~0.3–0.4 W; short stall peaks (~13 A) are fine with copper under the drain.

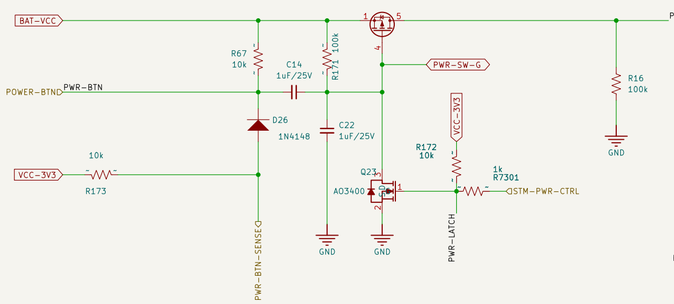
Old VM-5V switch removed: the LiDAR (VM-5V-LIDAR) and the water pump (VM-5V-WATER-PUMP) now have their own WD_OK-gated switches on their sheets.
No soft ramp.

VM-VBAT (motors, fan) is on only while WD_OK is high (RC watchdog fed) AND VM-VBAT-EN is low.
Q19 source-switched AND: GPIO low = on. R7010 keeps it off in reset/flash. AO4407C: Vgs +/−25 V, no gate divider needed.

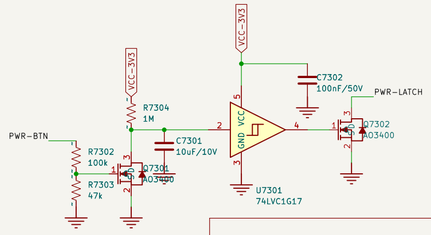




VUS = 0.6 V x (1 + 232k/10k) = 14.4 V



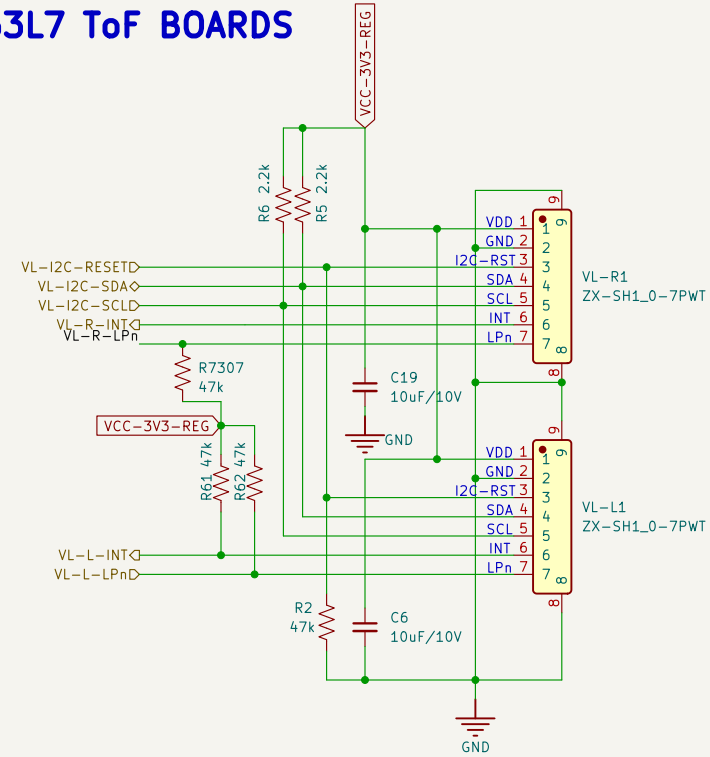
STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board



Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q7301 releases C7301, U7301 trips, Q7302 pulls PWR-LATCH low). Firmware turns off at 3 s.
While charging, the rail-hold (Q7303 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q7303 pulls PWR-SW-G low).

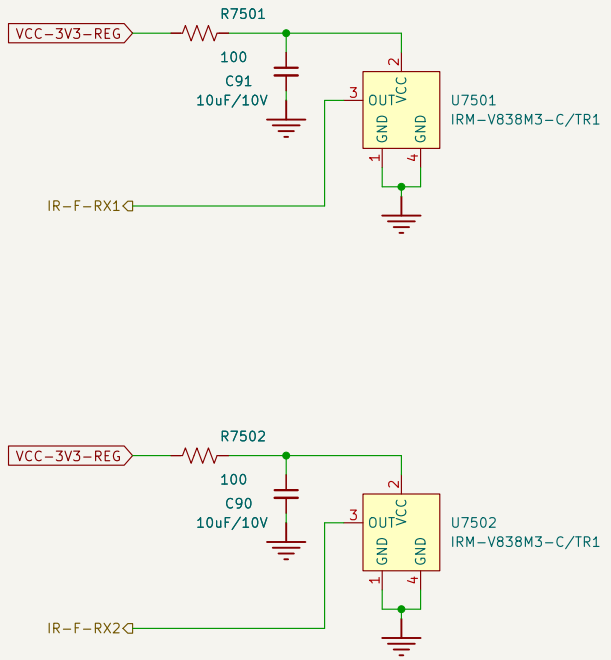
Sheet: /POWER/		
File: SYSTEM-POWER.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
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2x VL53L7 ToF BOARDS



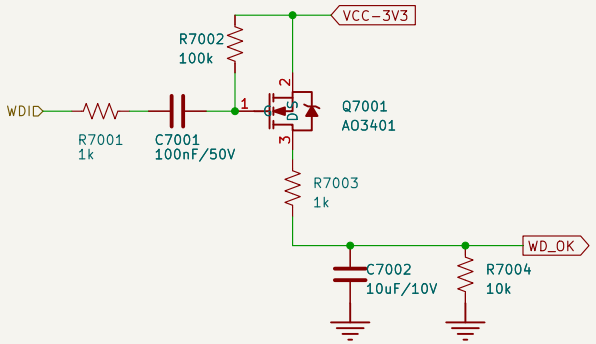
2 IR RECEIVERS

U7501 left, U7502 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window.



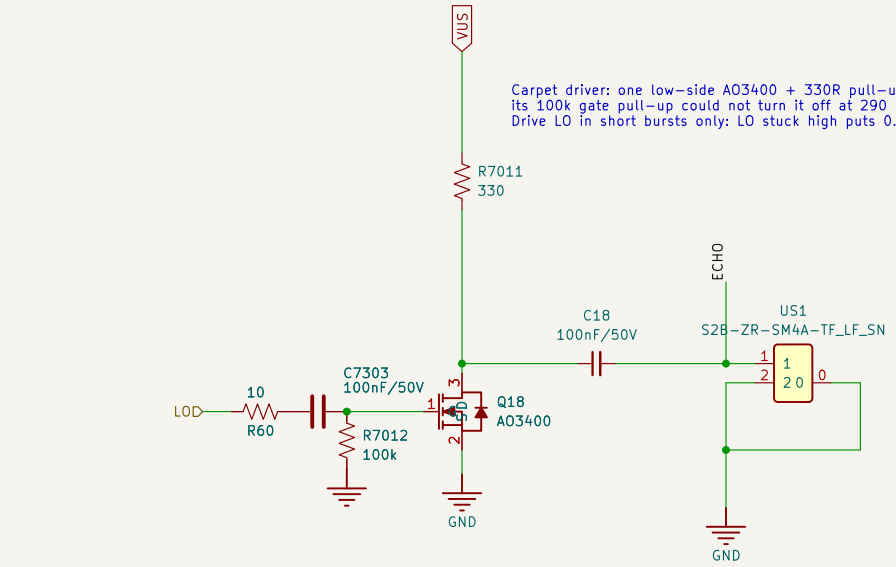
Sheet: /FRONT SENSORS/		
File: Front-Sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/16

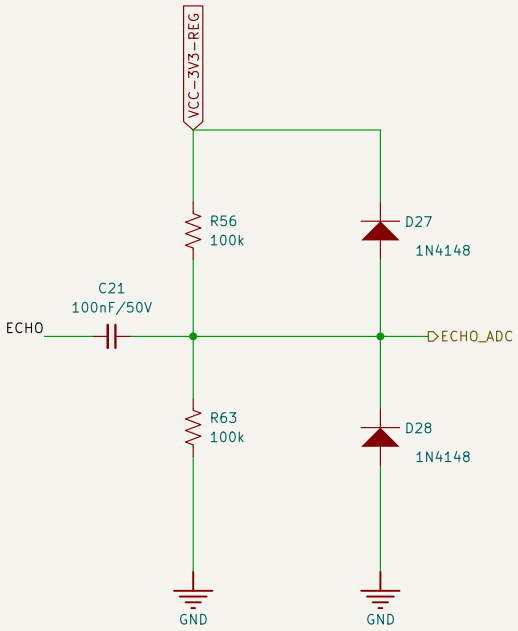
RC CHARGE-PUMP WATCHDOG: WD_OK gates VM-VBAT, VM-5V-LIDAR and VM-5V-WATER-PUMP



Firmware toggles WDI in software (50–1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q7001 gate low through C7001 and tops up C7002 via R7003.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

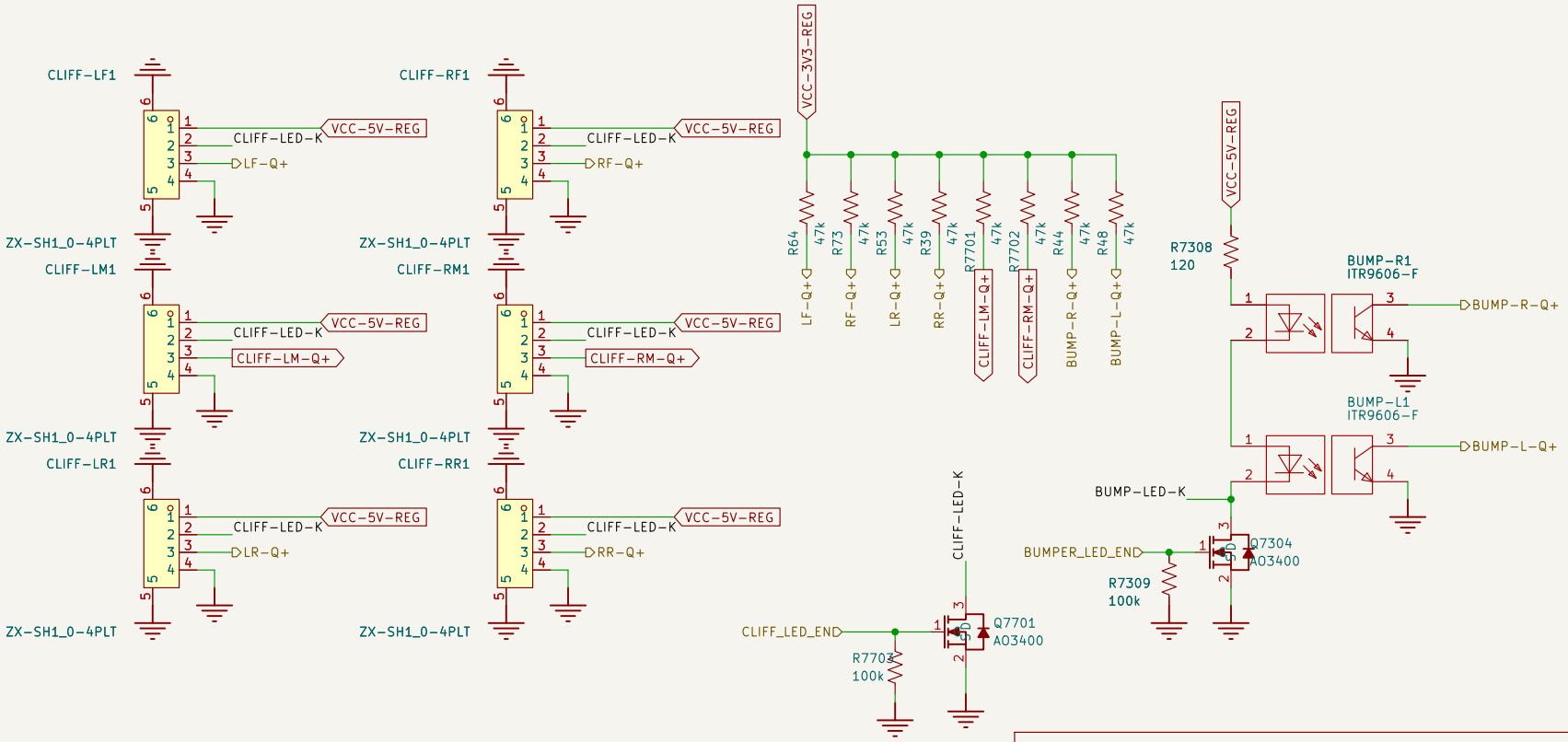
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File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 5/16





Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/16

ANTI FALL SENSORs

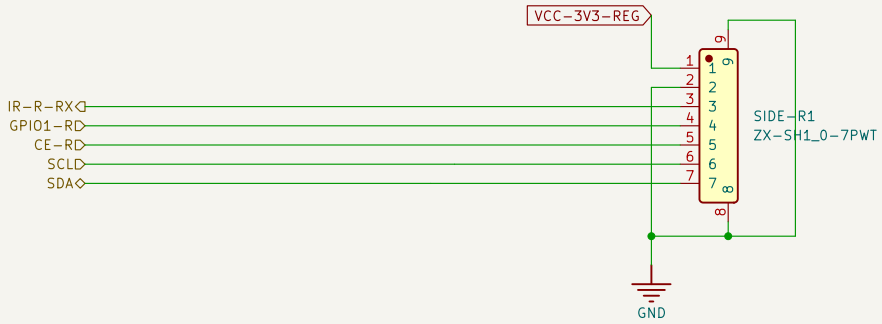


Cliff sensors: 6 boards (kicad/cliff-sensor), SH 4-pin vertical: 1 VCC-5V-REG (LED, 150R on the board), 2 CLIFF-LED-K (Q7701 sinks all 6 LEDs), 3 phototransistor collector (47k pull-up), 4 GND.
Modulate CLIFF_LED_EN and read each channel LED-on minus LED-off to cancel ambient light.
ADC: LF PA3, RF PA1, LR PA2, RR PA0, LM PD9, RM PE8 (LM/RM via global labels).

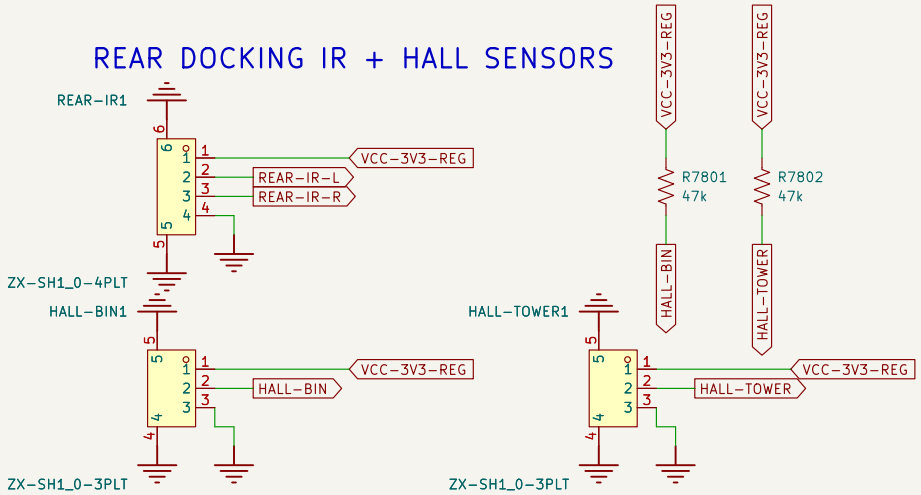
Bumpers: 2x ITR9606 slotted photo-interrupters on the main board (~11 mm tall, set by the bumper mechanism). Emitters in series from VCC-5V-REG, switched by BUMPER_LED_EN (high = on, 100k pull-down).
BUMP-x-Q+ reads low when light passes the slot (R44/R48 47k pull-ups).

Sheet: /CLIFF/	
File: CLIFF.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 19/16

SIDE PROXI IR SENSORs



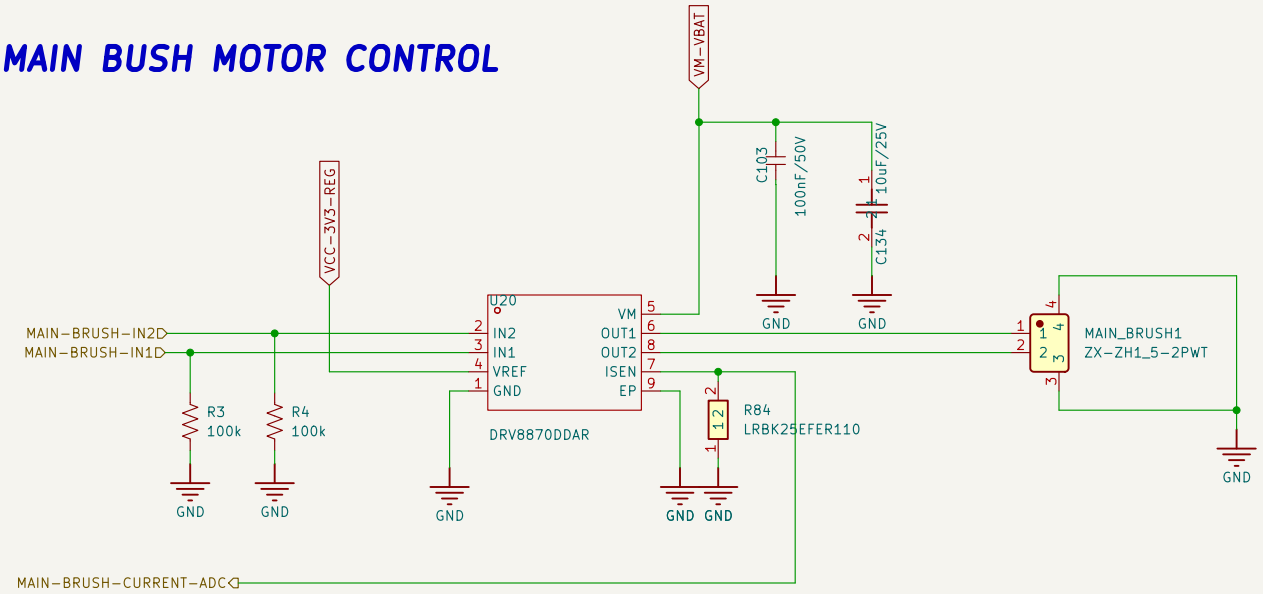
REAR DOCKING IR + HALL SENSORS



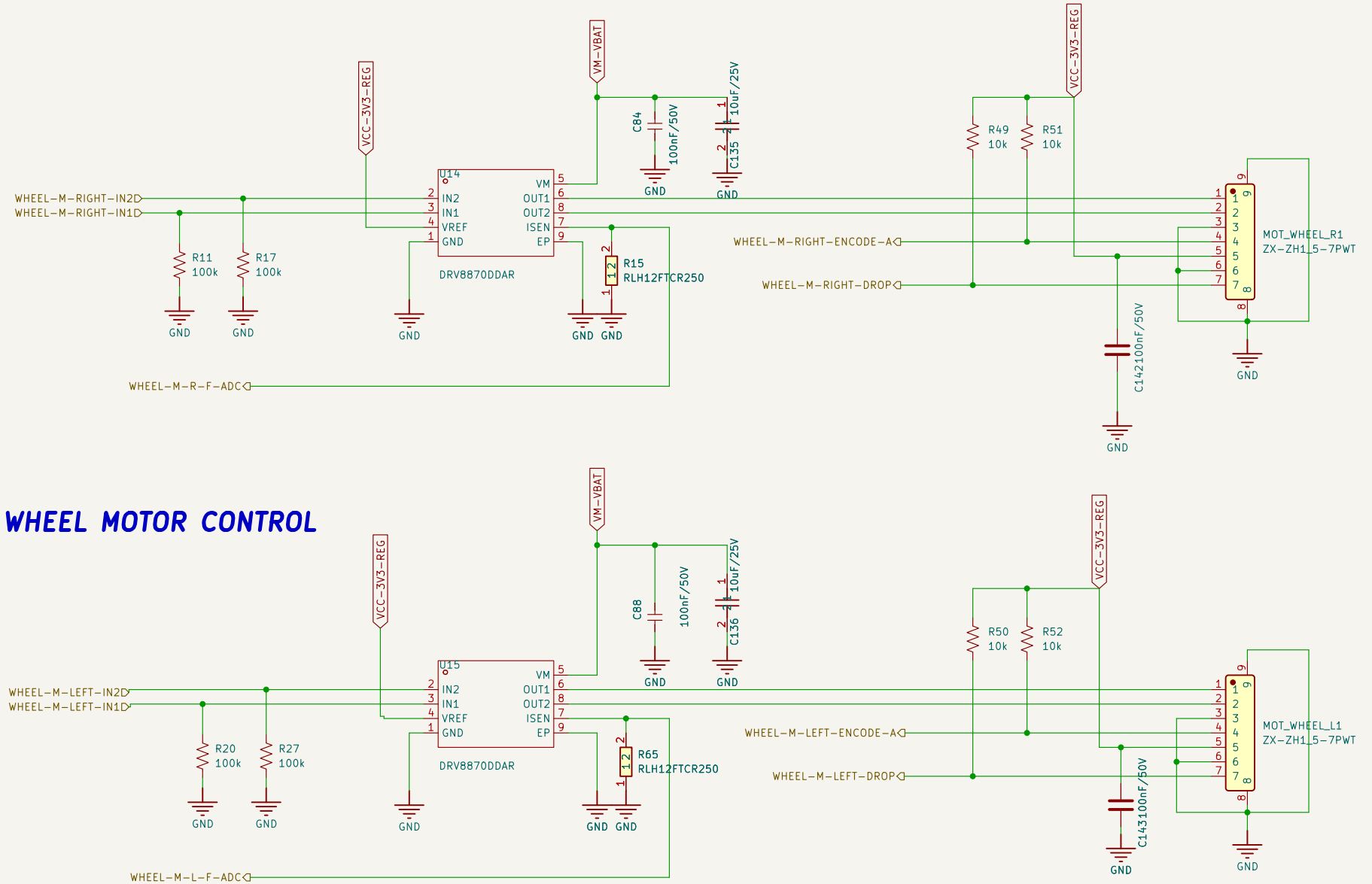
REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

Sheet: /SIDE-PROXI/	
File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 20/16

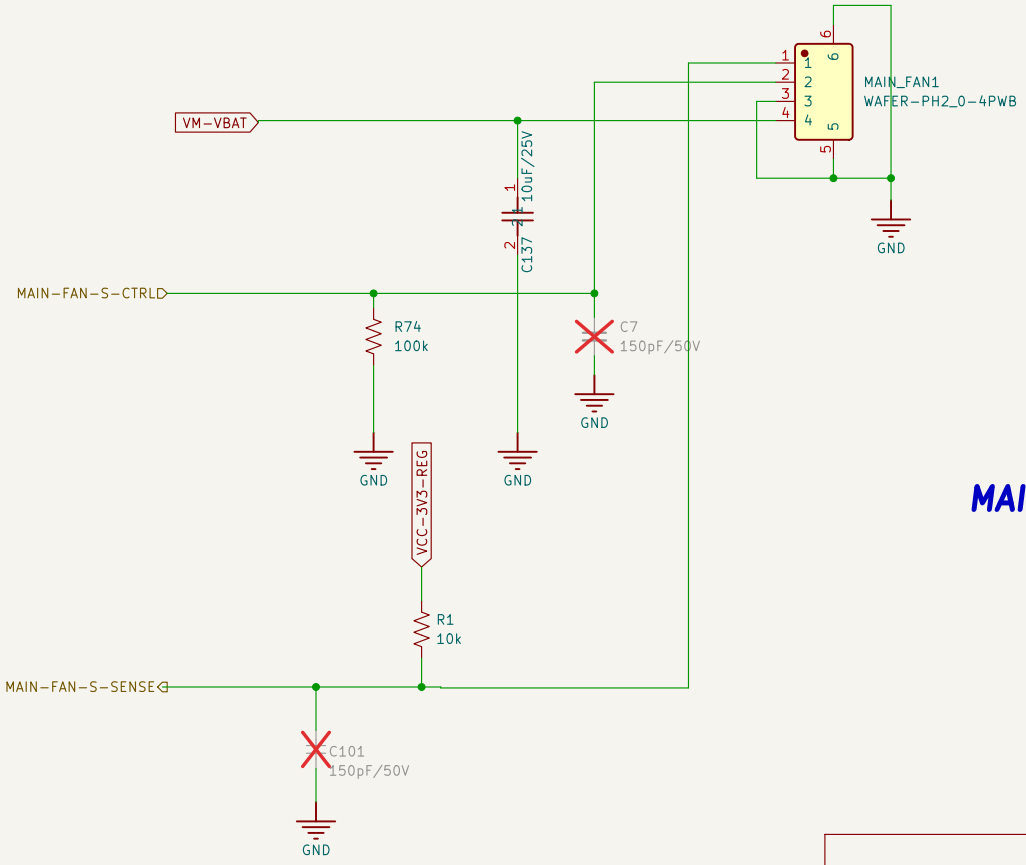
MAIN BUSH MOTOR CONTROL



Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/16



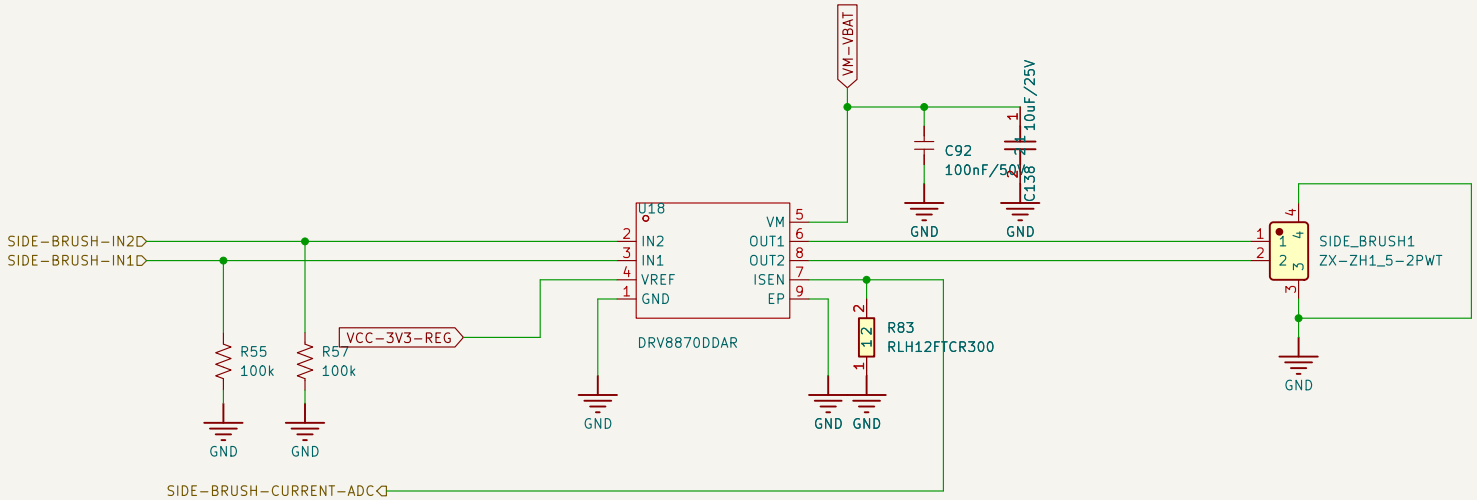
Sheet: /WHEEL MOTOR/	
File: WHEEL-MOTORs .kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 23/16



MAIN FAN CONTROL

Sheet: /MAIN-FAN/	
File: MAIN-FAN .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 24/16

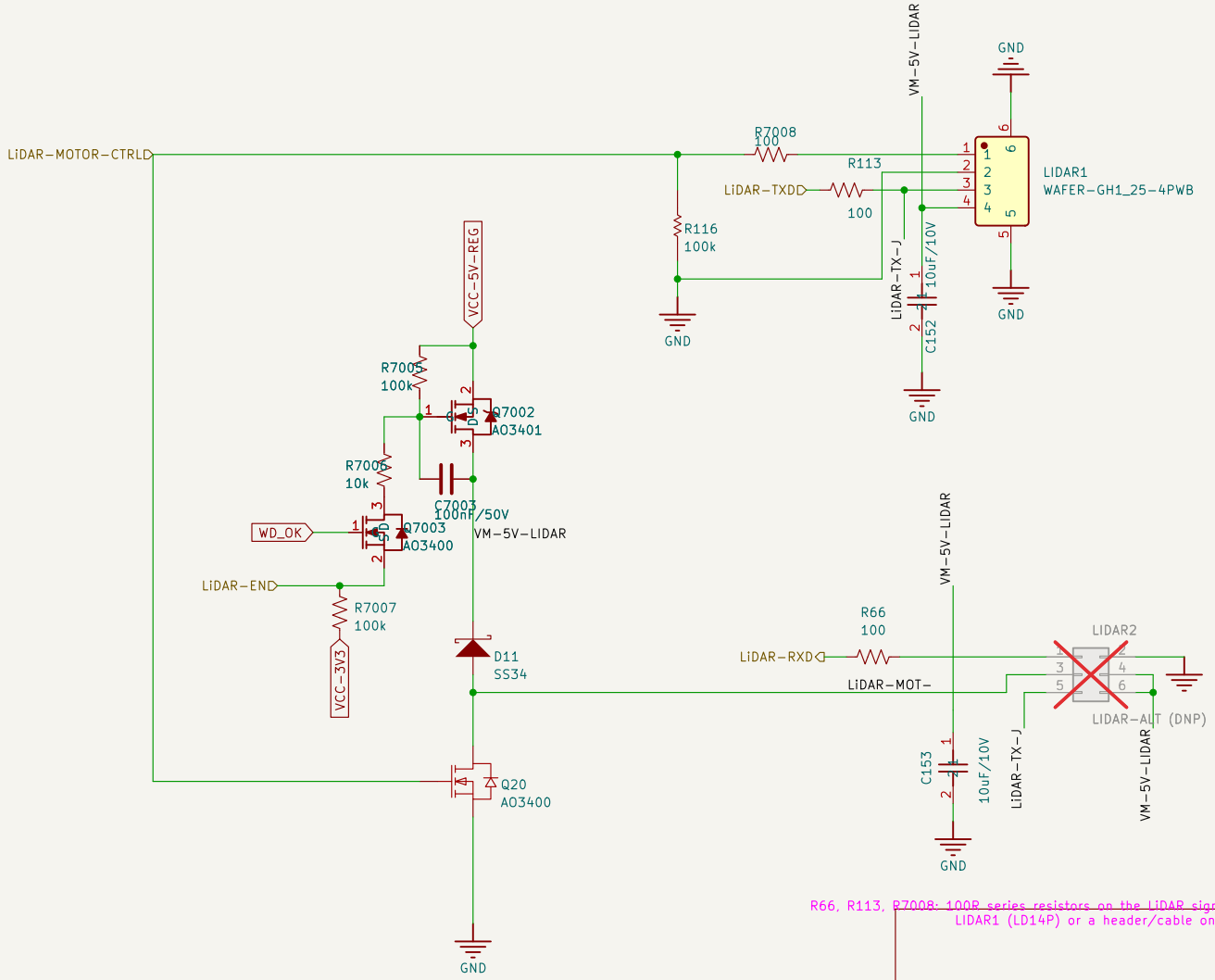
SIDE-BRUSH MOTOR CONTROL



Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 25/16

VM−5V−LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR−EN is low (Q7003 source−switched AND). R7006/C7003: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 2x3 0.1in through−hole pads (not fitted) for other LiDARs. Pin 6 = VM−5V−LIDAR for LiDARs
that need a separate motor supply (motor between pins 6 and 3). Pin 5 = the LiDAR TX, shared with LIDAR1 pin 3.

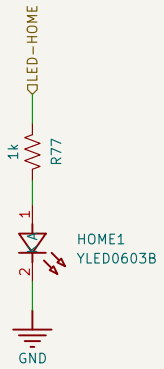
LiDAR

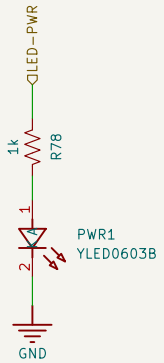


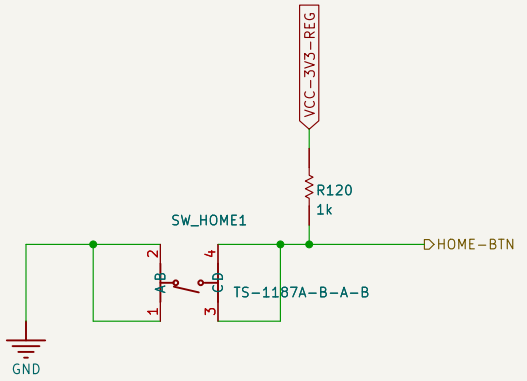
R66, R113, R7008: 100R series resistors on the LiDAR signals. Only one LiDAR is fitted:
LIDAR1 (LD14P) or a header/cable on LIDAR2.

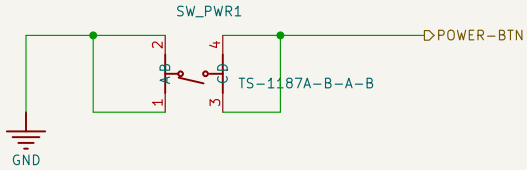
Sheet: /LiDAR/ File: LiDAR.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/16

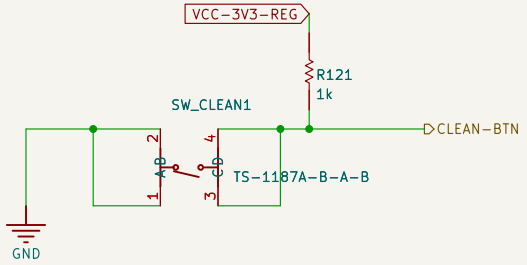
LEDs AND BUTTONs







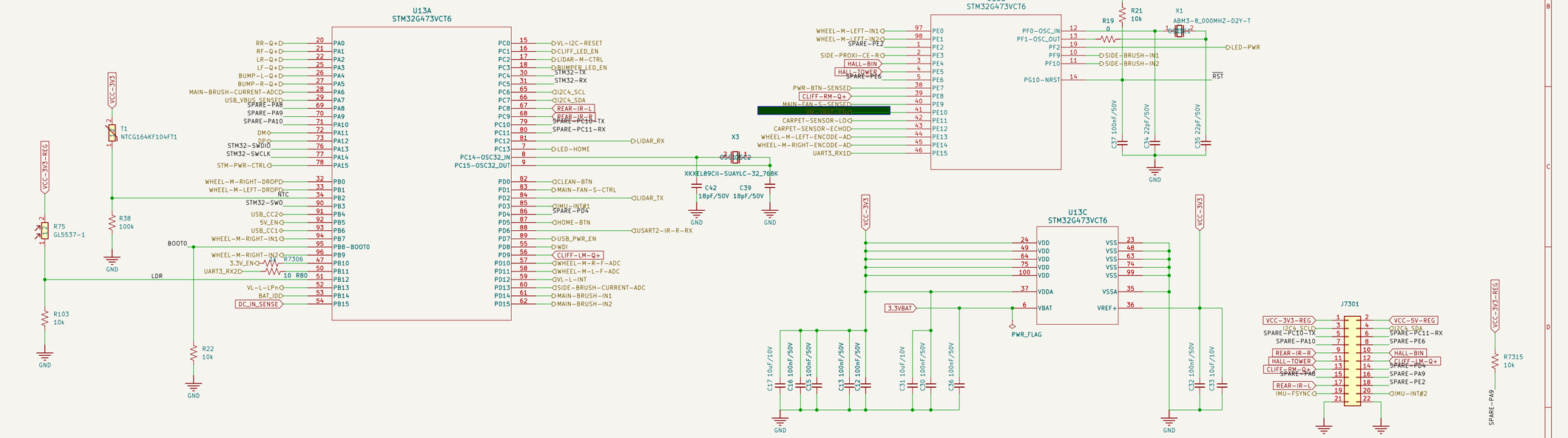


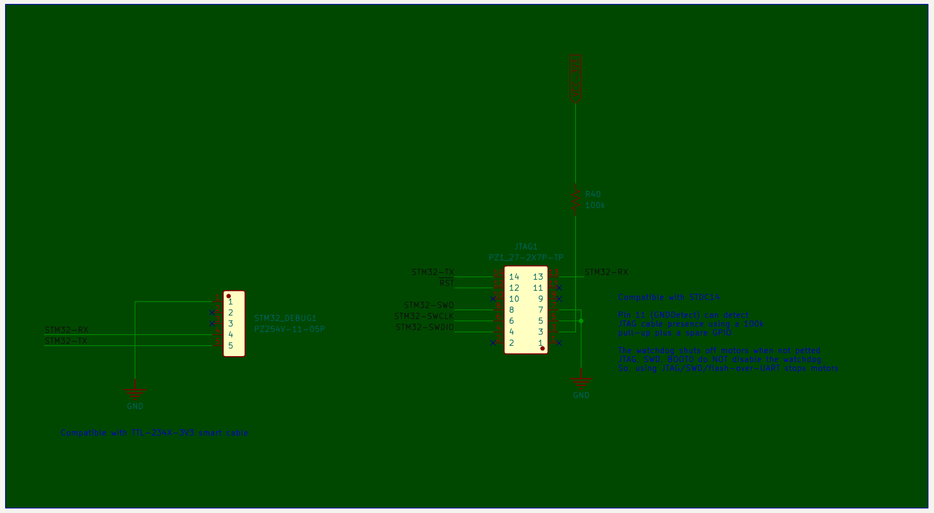


Sheet: /BUTTON-LED/		
File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 28/16

MCU STM32G473

Phone fork: PB6/PB4 = UCPD1 CC1/CC2 (USB-C). PB5 = 5V_EN. PA7 = USB_VBUS_SENSE (ADC2_IN4).
Front ToF moved from the PI: I2C on I2C4 (shared with the side boards). PC0 = VL-I2C-RESET.
PD12 = VL-L-INT. PC8 = VL-R-INT. PB13 = VL-L-LPn. PE8 = VL-R-LPn. Use internal pull-ups on INT.





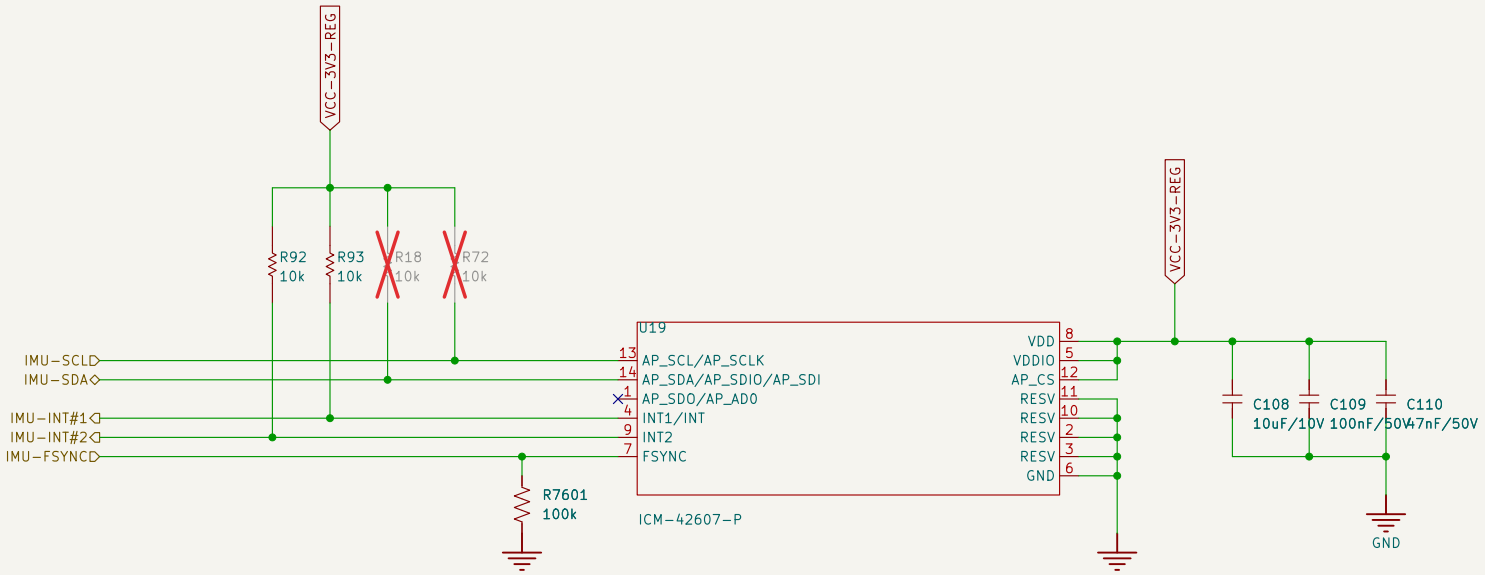
SPARE-IO (DNP)

J7301 spare I/O: 0.1in 2x11 pads, do not fit (zero height).
1 VCC-3V3-REG (sensor rail), 2 VCC-5V-REG, 21/22 GND.
3/4 I2C4 SCL/SDA (shared with the ToF, side sensor and IMU; 2.2k pull-ups).
5/6 UART PC10 TX / PC11 RX (UART4 or USART3). 7-18 spare GPIO, except
9/17 rear IR R/L, 10/11 Hall bin/tower, 12/13 cliff LM/RM. 19/20 IMU FSYNC/INT2 (FSYNC: 100k pull-down on the IMU sheet). SWO (PB3) on JTAG1 is a spare when not tracing.

R7315: PA9 = UCPD DBCC1. High during reset
turns on the dead-battery Rd on CC1 (PB6).

Sheet: /MCU-STM32/	
File: STM32G473.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 30/16

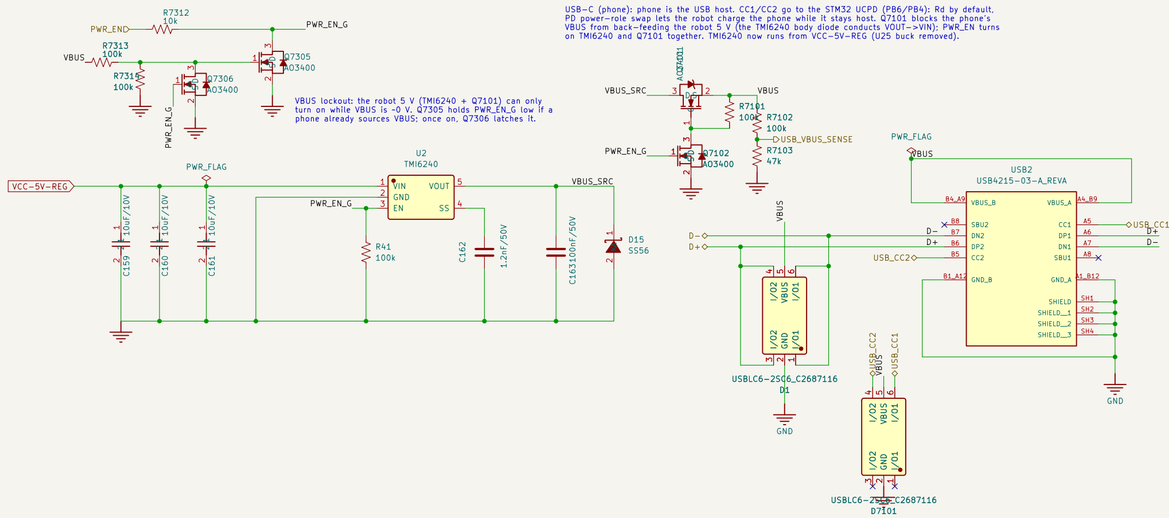
IMU-ICM-42607-P



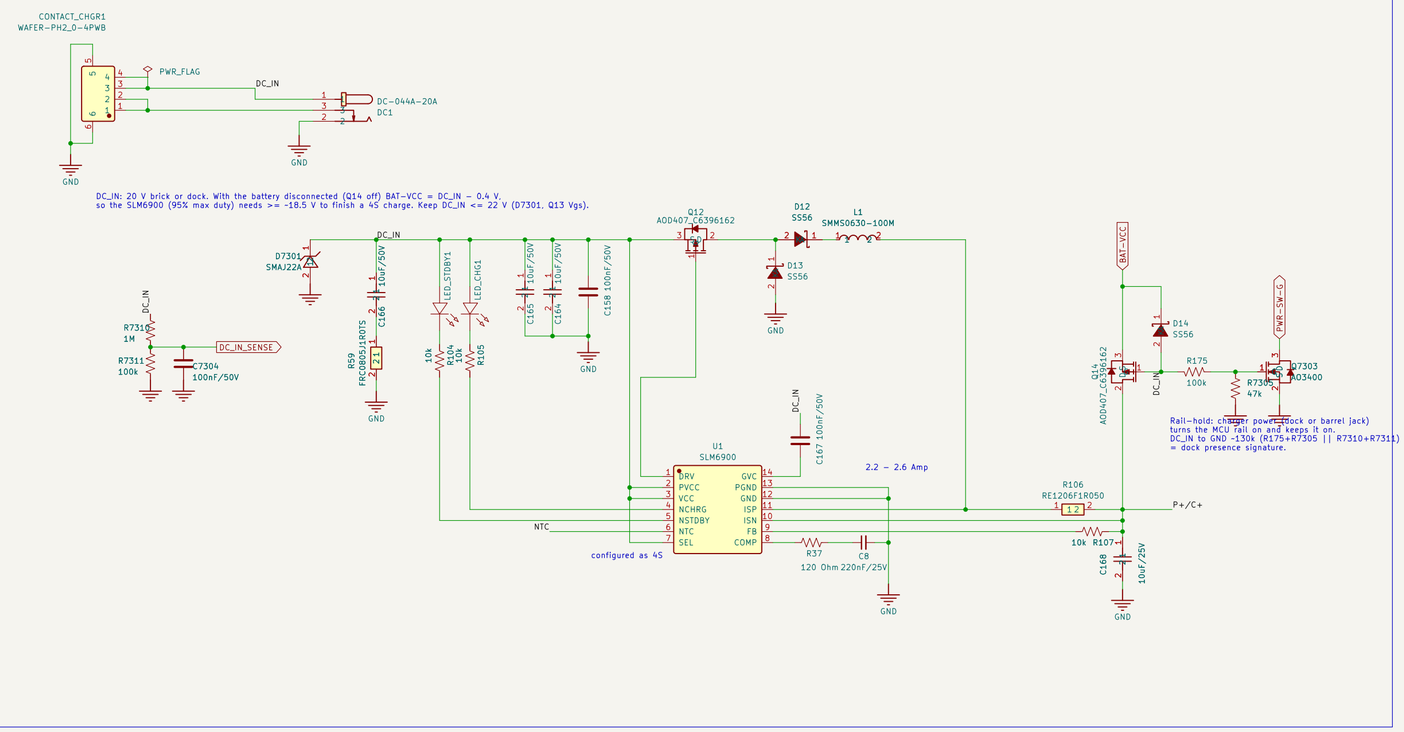
R18/R72: not fitted. The IMU is on I2C4 now (2.2k pull-ups on the MCU sheet).

Sheet: /IMU/	
File: IMU-ICM-4267-P .kicad_sch	
Title:	
Size: USLetter	Date:
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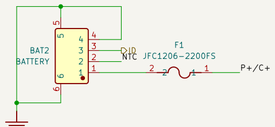
USB-C



BATTERY CHARGER



BATTERY CONNECTOR



Sheet: /BATTERY-CHARGER/	
File: Battery-Charger.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.4	**Rev:** Id: 33/16